



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION II

CANDIDATE
NAME

CENTRE
NUMBER

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INDEX
NUMBER

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CLASS

H2 BIOLOGY

9648/03

Applications Paper and Planning Question

22 September 2015

Paper 3

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	/ 14
2	/ 11
3	/ 15
4 (SPA)	/ 12
5	/ 20
Total	

This Question Paper consists of **16** printed pages.

Answer **all** questions.

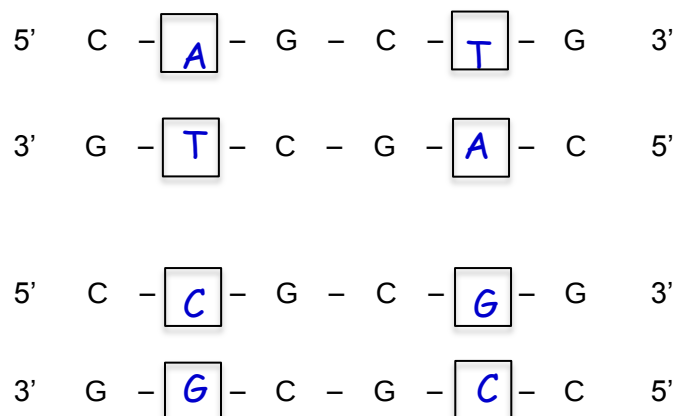
1. Restriction enzymes are widely used in genetic engineering, including the production of DNA library.

Table 1.1 shows the restriction site recognised by two restriction enzymes. **K** may be one of the bases: G or T and **M** may be one of the bases: A or C.

Table 1.1

Restriction enzyme	Cut sites within the restriction site
<i>Sa</i> I	5' G T C G A C 3' 3' C A G C T G 5' (Cuts between T and C on top strand, and between G and C on bottom strand)
<i>MspA</i> II	5' C M G C K G 3' 3' G K C G M C 5' (Cuts between M and G on top strand, and between K and C on bottom strand)

- (a) (i) Complete Fig. 1.1 to show two possible arrangements of bases of the restriction site of *MspA*II [2]



1 mark for each correct sequence

Fig. 1.1

- (ii) With reference to Table 1.1, suggest why *MspA*II is not used in the formation of recombinant DNA. [3]

- MspA*II makes a double-stranded cut in the middle of the restriction site, resulting in a blunt ends made annealing more difficult;;
- additional steps required to add linkers for hydrogen bond to form;;
- RE cut at more than one restriction site therefore less specific / RE could cut at the wrong place and therefore the insert the gene in the incorrect location;;

- (b) For the production of human insulin using prokaryotic cells, the gene for insulin can be isolated from either cDNA library or genomic DNA library.

Explain the advantages of obtaining insulin gene from a cDNA library instead of a genomic DNA library.

[3]

cDNA library	Genomic DNA library
Contains only coding regions / does not contain introns → continuous coding strand;	Contains both coding and non-coding sequence / introns are present;
Does not required, RNA processing/remove introns/RNA splicing;	Requires, RNA processing/remove introns/RNA splicing, which is absent in prokaryotic cells;
The gene is <u>intact</u> ;	Gene may be <u>fragmented</u> ;
Easier to find the gene since <u>less clones to screen</u> ;	More difficult to find the gene as <u>more clones to screen</u> ;

- (c) Describe how bacterial colonies that contain insulin gene can be identified in cDNA library.

[2]

1. Colonies transferred to nitrocellulose membrane;
2. DNA is denatured to form single-stranded DNA;
3. Radioactively labelled, single-stranded DNA probe;
4. With sequence complementary to insulin gene;
5. Hybridise via complementary base pairing / hydrogen bonds;
6. Form an image on the X-ray film;
7. Correspond to the colony that contains the insulin gene;

- (d) Once the insulin cDNA has been isolated, polymerase chain reaction can be used to amplify the gene. An enzyme, Taq DNA polymerase, was extracted from a bacterium, *Thermus aquaticus*, which is found in the hot spring of the Yellowstone National Park in the USA.

Explain the advantage of the use of Taq DNA polymerase in the polymerase chain reaction.

[2]

1. resistant to denaturation by heat treatment;;
2. Can be used repeatedly / the enzyme need not be replaced at the end of every cycle;

Insulin is a chain of 52 amino acids. Insulin gene can be synthesised *in vitro*. Fig. 1.2 summarises the production of human insulin and human factor VIII.

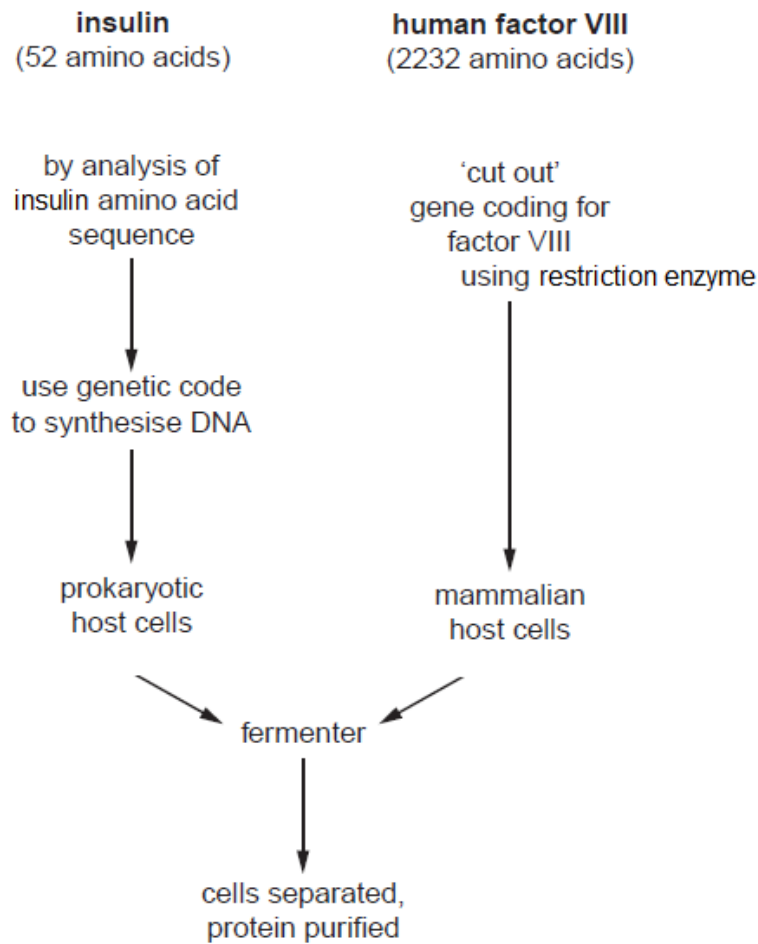


Fig. 1.2

- (e) Suggest why the method used to synthesise the gene coding for human insulin is not used to obtain the gene coding for human factor VIII. [1]

Any one

- Polypeptide / protein, too long/ large, to sequence;;
- amino acid sequence not, fully determined;;
- known nucleotide sequence too long for chemical synthesis;;
- Easier / cheaper, to use restriction enzymes;;

[Total: 14]

2. Children with severe combined immunodeficiency disorder (SCID) cannot produce many types of white blood cells that are involved in immune response. This is because they do not have the functional gene to make the enzyme adenosine deaminase (ADA).

Some children with SCID have been treated with stem cells.

The treatment used is described in Fig. 2.1.

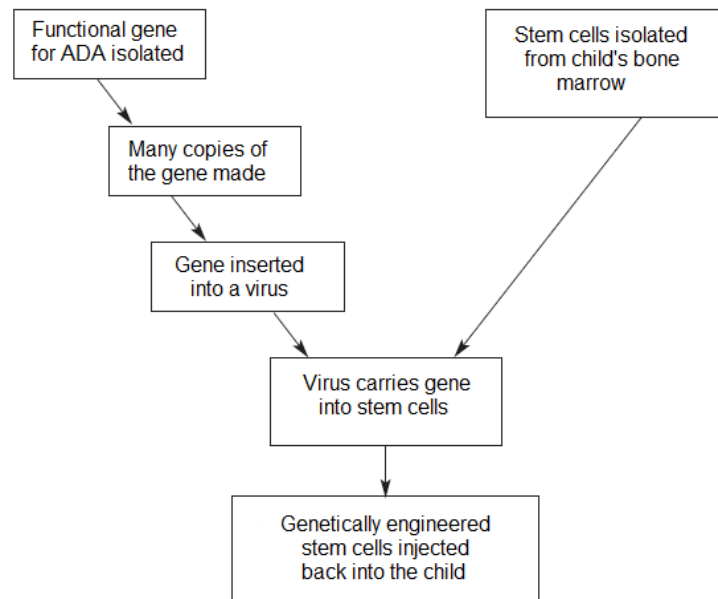


Fig. 2.1

- (a) Explain how ADA-deficiency could lead to SCID.

[2]

1. ADA involved in metabolism of nucleotide deoxyadenosine triphosphate (dATP) / lack of enzyme results in accumulation of dATP in T lymphocytes;
2. killing the T lymphocytes;
3. without T lymphocytes, B lymphocytes cannot recognise antigen and produce antibodies;
4. weaken immune systems so susceptible to microbial infections;

- (b) Using the information given, suggest and explain two reasons why stem cells were used in this treatment.

[2]

Any two

1. they can divide and grow indefinitely; hence, they can replenish their own population / can replace existing (faulty) cells;
2. they can differentiate into specialised cells under appropriate conditions; hence, produce lymphocytes with the normal functional allele to fight infection;
3. using child's own cells; hence no/little risk of rejection;

Other than viral delivery method, liposomes have been used to deliver functional ADA gene into stem cells *in vitro*. In both cases the cells were injected into the patients after modification.

Fig. 2.2 shows the number of functioning white blood cells in the child during the year following treatment. Children who do not suffer from SCID have between 5000 and 8000 white blood cells per mm^3 of blood.

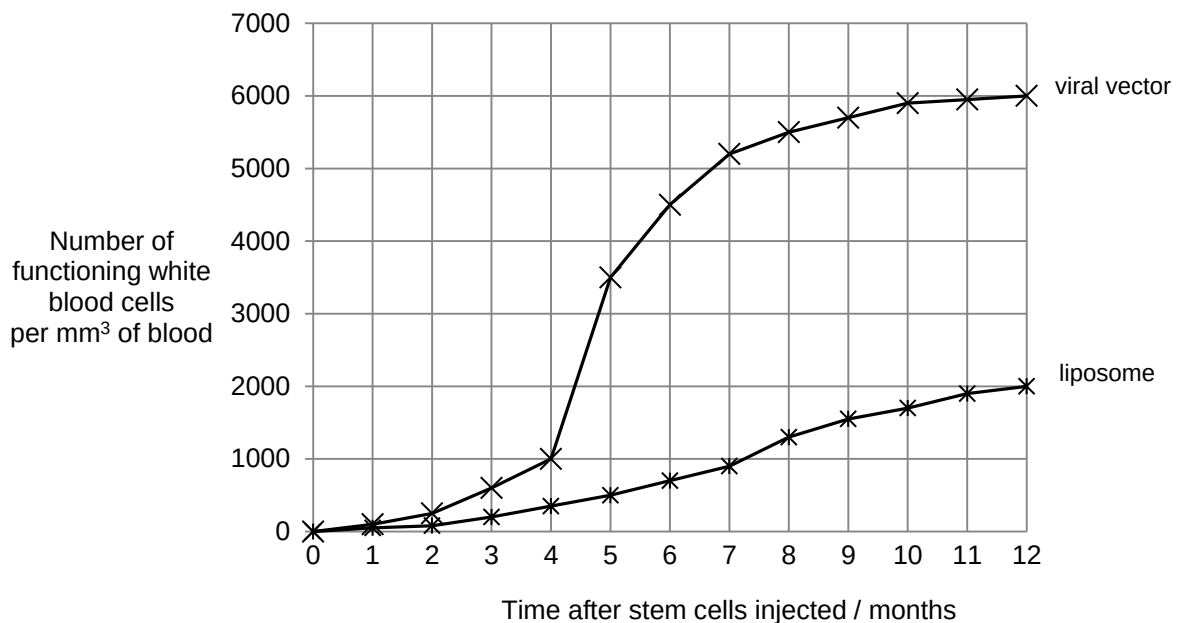


Fig. 2.2

(c) Explain how liposome is able to deliver the normal gene into the stem cells. [2]

1. Liposomes are composed of synthetic lipid bilayers which mimic the structure of biological molecules;;
2. Allowing fusion with the plasma membrane of stem cell;;

(d) With reference to Fig. 2.2, describe and explain the results. [3]

From 0 to 12 months,

1. Viral method: the number of functioning white blood cells increases from 0 to 6000 cells per mm^3 of blood;
2. Liposome: the number of functioning white blood cells increases gradually from 0 to 2000 cells per mm^3 of blood;

Viral vector	Liposome
1. Targeted gene transfer; more cells take up the ADA allele to have an effect;	1. Non-specific; No enough cells take up the ADA allele to have an effect;
2. Allele integrated into host cell's	2. Allele not integrated into host

chromosomes; conferring long term stability / not degraded / can pass to daughter cells / gene will likely be expressed ;	cell's chromosome; degraded by cellular enzymes / cannot pass to daughter cells / gene is not expressed;
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(e) State one ethical and one social concern regarding the use of gene therapy. [2]

Ethical: *any one*

- Altering ones genes and changing the way one was created goes against many religious and cultural beliefs;;
- Germ-line gene therapy would affect the human gene pool permanently and any error in technology or judgment could have far-reaching consequences;;
- Gene therapy may be used to enhance genetic profile of individuals and who decides if it is gene enhancement or treatment?;;

Social: *any one*

- Differential access to gene therapies could possibly lead to disparity between the rich and the poor, with the rich having better accessibility to treatments;;
- Some also fear that widespread use of this technology could lead to new definitions of "normal" with regards to human beings, less accepting of people who are different;;

[Total: 11]

3. In humans, iron-deficient anaemia is one of the most serious problems worldwide.

Transgenic varieties of rice are produced to address the problem of iron deficiency. Increased expression of the Nicotianamine Synthase (NAS), an iron chelator, is found to increase iron content in transgenic rice plant. In order to elevate expression of the NAS gene in the rice genome, a more active control element is inserted upstream of the NAS gene.

The most common vector for introducing new DNA sequences into plant cells is a naturally occurring Ti-plasmid derived from *Agrobacterium tumefaciens*, shown in Fig 3.1.

This plasmid integrates a segment of its DNA known as T-DNA into the chromosomal DNA of host plant cells following infection. Expression of genes in the T-DNA region of the plasmid, in plants cells, leads to crown gall disease.

To produce transgenic rice, the isolated Ti plasmid is engineered and reintroduced back into *A. tumefaciens*. It is then allowed to infect the plant.

The process showing how transgenic rice plant is produced is illustrated in Fig 3.2.

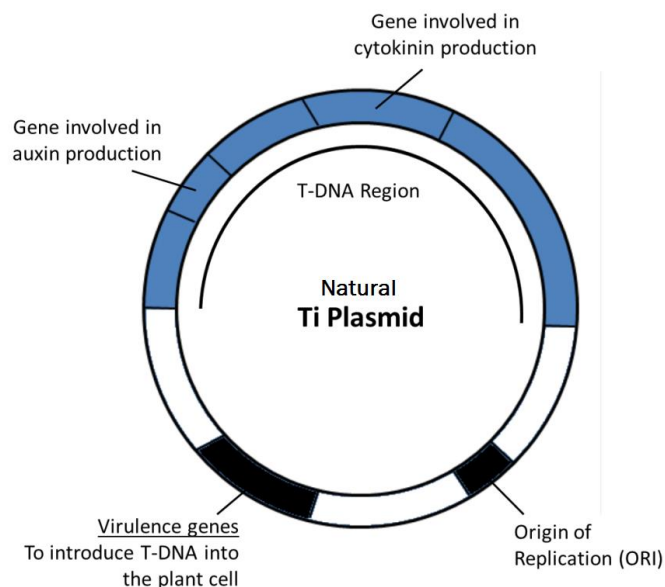


Fig. 3.1

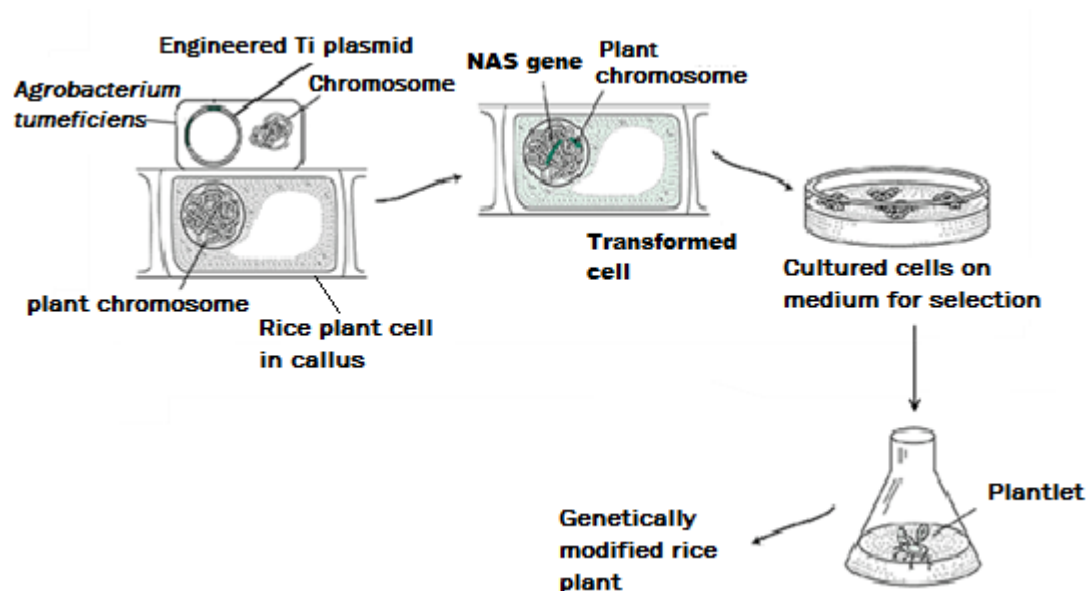


Fig 3.2

- (a) Using information from Fig 3.1 and Fig 3.2, describe and explain two modifications to the Ti plasmid before it can be used in the production of genetically modified rice producing elevated levels of NAS protein. [4]

Any 2 pairs

1. Insert more active control element to the T-DNA region of the plasmid;;
2. Allow integration of control element into the plant genome and hence NAS gene expression;;
3. Insert genetic/selection markers along with the desired gene to T-DNA;;
4. Required for the selection of plant cells with the T-DNA / transformed plant cells;;
5. Genes on T-DNA that codes for protein involved in the synthesis of auxin and cytokinin should be removed;;
6. Prevents formation of gall (abnormal outgrowth)

- (b) Outline how transformed cells are developed into plantlets. [5]

1. Cells were grown in culture vessels containing nutrients and plant growth regulators;;
2. Auxin/cytokinin is added to the nutrient agar to stimulate the cells to divide by mitosis forming a callus;;
3. The number of calli increased by subculturing;;
4. The cells of the callus are induced by auxin and cytokinin/gibberellin to differentiate into a plantlet;;
5. Low level of cytokinin and high level of auxin triggers formation of roots growth while high level of cytokinin and low level auxin triggers shoot growth;;

- (c) The haemoglobin count of ten iron-deficient girls, aged thirteen, were analysed. Their haemoglobin count is below 12.5 g dl^{-3} .

Subject 1 to 8 adhered to a diet of transgenic rice with increased iron content, three times a day, for three months.

Subject 9 and 10, had a diet consisting of normal rice grown in the same rice field, three times a day, for three months.

The normal haemoglobin count of girls at the age of thirteen is within the range of $12.5 - 16.0 \text{ g dl}^{-3}$.

The results of the analysis are shown in Table 3.1.

Table 3.1

	Subject	Haemoglobin count/ g dl^{-3}
Diet consisting of transgenic rice with increased iron content	1	14.0
	2	16.0
	3	12.4
	4	14.5
	5	12.3
	6	14.1
	7	22.1
	8	13.0
Diet consisting of normal rice	9	11.2
	10	10.9

- (i) Using the data provided in Table 3.1, comment on how successful is transgenic rice in addressing iron deficiency. [2]

1. Ref to subject 9 and 10, all subjects with diet consisting of transgenic rice have an increased in iron content;
2. Quote value (for either individuals achieving normal haemoglobin count/remain below normal haemoglobin count or excess haemoglobin count);
3. Ref to different extent of success between individuals;
4. However, individual 7 may have suffered from iron overdose;

- (ii) What is a possible social implication that may arise from genetic engineering of plants? [1]

Any one

- potential of human health impact such as allergens which some humans are allergic to, while consuming plant as food source;;
- threat to human safety as transfer of antibiotic resistance gene from the plasmid to bacteria in the gut;;

Other than genetically engineering plants to produce more nutritious food, genetically engineered organisms can also be used in solving the world demand for food.

- (d) Describe how one such organism is genetically modified to mitigate food shortage. [3]

Any one

1. GM salmon;;
 2. Insertion of promoter sequence from the ocean pout;
 3. Which is active all year round;
 4. And not only in the presence of light;
 5. Enabling GM salmon to grow continuous;
 6. Salmon ready for sale in a much shorter time, increasing food supply;;
1. Bt crop;;
 2. Bt toxin gene from soil bacterium *Bacillus thuringiensis*;
 3. encodes insecticidal crystalised proteins (ICPs);
 4. ICPs bind to receptors on intestinal lining, rupture cells;
 5. Genetically modified corn produces ICP, killing larvae of *Lepidoptera* species;
 6. larvae are unable to cause damage to crops hence increasing crop yield;;
1. Favr Savr Tomato;;
 2. insert the antisense polygalacturonase (PG) gene into the tomato;
 3. antisense mRNA of PG hybridise with PG mRNA;
 4. via complementary base pairing;
 5. the process of translation is blocked;
 6. delayed ripening genes which increases the shelf life of the tomato/ can be freight over longer distances for distribution;;

[Total: 15]

4. Planning Question

Lipase is an enzyme the body uses to break down triglycerides in food so they can be absorbed in the intestines. Lipase catalyses the hydrolysis of triglycerides into fatty acids and glycerol. The end point of this hydrolysis can be determined by using an indicator which changes from colourless in alkaline pH to yellow when the fatty acids are produced.

Dietary fats do not dissolve in water, and bile salts secreted by the liver are required to emulsify fats. This provides a larger surface area for lipase to act on.

Orlistat, a weight-loss agent with a novel mechanism of action, was recently approved by the Food and Drug Administration for the treatment of obesity. It inhibits gastric and pancreatic lipases in the lumen of the gastrointestinal tract to decrease systemic absorption of dietary triglyceride.

Using this information and your own knowledge, design an experiment to determine if Orlistat functions as a competitive inhibitor or a non-competitive inhibitor of lipase.

You must use:

- milk, the triglyceride content of the milk is 5 %,
- distilled water,
- 1 % lipase,
- 25 μ M Orlistat,
- indicator,
- alkaline solution,
- bile salt solution.

You may select from the following apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods etc.,
- syringes,
- pipette filler,
- white tile,
- bunsen burner with tripod, gauze and bench mat,
- timer e.g. stopwatch or stop clock,
- thermometer.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

Reaction/Process [P= max 1]

1. Lipase catalyses the hydrolysis of triglyceride into fatty acid and glycerol
2. Lipase has a specific active site/ ref to lock- & key hypothesis;
3. Triglyceride fits into the active site to form an E-S complex;
4. Resulting in the breakage of the ester linkages;
5. Orlistat inhibit lipase, prevent the binding of triglyceride to the active site, the rate decreases;

Method of measurement [M= max 1]

6. Fatty acids product of the fats digestion will cause a change in pH which will make the indicator changing from colourless to yellow;
7. The time taken for the indicator to turn yellow is inversely proportional to the rate of hydrolysis of triglyceride to fatty acid and glycerol;

Expected trend

Competitive inhibition: [CI= max 1]

8. Inhibitor is structurally similar to the substrate;
9. Binds reversibly to the active site of the enzyme;
10. Competes with the substrate for active site;
11. Effect of inhibition can be overcome by increasing substrate concentration;

Non-competitive inhibition: [NCI= max 1]

12. Inhibitor binds to site other than active site of the enzyme;
13. Change formation of active site;
14. Substrate cannot bind to active site;
15. Inhibition cannot be overcome by increasing substrate concentration;

Experimental variables [V= max 2]

16. Independent variable : a. Concentration of triglyceride;
b. Range: 1 to 5 %;
17. Dependent variable : Concentration of fatty acid formed in term of time taken to reach end point;
18. Controlled variables :

	Controlled Variable	Appropriate Quantity
a	volume of triglyceride (i.e. milk) used	5 cm ³ ;
b	Volume of lipase used	2 cm ³ ;
c	volume of Orlistat used	1 cm ³ ;
d	Volume of bile salt solution / alkaline solution	1 cm ³ ;
e	Volume of indicator	0.5 cm ³ ;
f	temperature	37°C;

At least 2 for $\frac{1}{2}$ mark

With inhibitor added

Procedure [P= max 2]

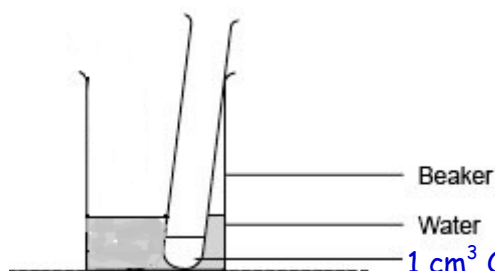
19. Dilute the 5% triglyceride stock to give 10 cm³ of each concentrations:

Dilution table of triglyceride [D= $\frac{1}{2}$]

Concentration of diluted triglyceride/ %	Volume of stock triglyceride/ cm ³	Volume of water/ cm ³
1	2.0	8.0
2	4.0	6.0
3	6.0	4.0
4	8.0	2.0
5	10.0	0.0

20. Add 5 cm³ of 1% triglyceride;
21. Add 1 cm³ of bile salt solution into the test tube labelled 1%;
22. Add 1 cm³ of alkaline solution;
23. Add 0.5 cm³ of indicator;
24. Add 1 cm³ of 25 μ M Orlistat;
25. Add 2 cm³ of lipase into the same test tube;
26. Mix and start the stopwatch;
27. Incubate the test tube in a water bath maintained at 37 °C;

28. Record the time taken for indicator to turn yellow using a stopwatch;
29. Record the reading in an appropriate table;
30. Repeat steps 21 - 29 for the rest of triglyceride concentrations: 2 %, 3 %, 4 % and 5%;



Water bath maintained at 37°C
Reject: direct heating

1 cm³ Orlistat 1 cm³ of bile salt solution
 5 cm³ milk 1 cm³ of alkaline solution
 2 cm³ lipase 0.5 cm³ of indicator

Without inhibitor added [$C = \frac{1}{2}$]

31. Repeat steps 19 to 30 except that 1 cm³ of Orlistat is replaced with equivalent volume of distilled water;

Replicates and repeats [$R \& R = \frac{1}{2}$]

32. ensure that there are 3 replicates for each concentration of triglyceride;
33. Repeat the entire experiment twice times;

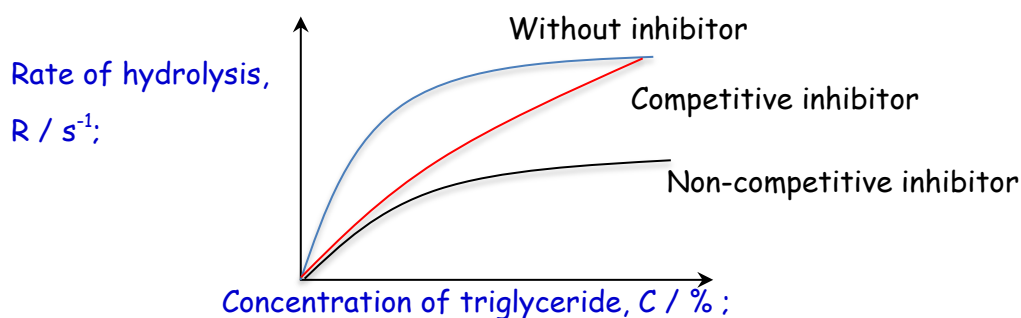
Data manipulation

Table [$T = 1$]

Rate = $1 / \bar{T}$

	With inhibitor					Without inhibitor				
Concentration of triglyceride, $C / \%$	34. Time taken to reach end point, T / s ;				35. Rate of hydrolysis, R / s^{-1}	Time taken to reach end point, T / s ;				Rate of hydrolysis, R / s^{-1}
	T_1	T_2	T_3	$\overline{T}$		T_1	T_2	T_3	$\overline{T}$	
1										
2										
3										
4										
5										
control										

Graph [G= 1]



36. Correct axes;

37. Correct shape of graphs - competitive & non-competitive inhibition;

Safety precautions [S=max 1]

38. Lipase and bile salt are skin irritant. Wear glove and goggles;

39. Alkaline solution is skin irritant. Wear glove and goggles;

40. AVP;;

5. Free Response Question

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate.
- must be in continuous prose, where appropriate.
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL** return is necessary if you have not attempted this section.

(a) Describe how human growth hormone is produced by genetic engineering. [7]

(b) Explain how RFLP analysis facilitates the process of detecting sickle cell anaemia. [8]

(c) Discuss the ethical concerns that arose from the human genome project. [5]

[Total: 20]

- (a) Describe how human growth hormone is produced by genetic engineering. [8]

Synthesis of recombinant plasmid carrying hGH gene

1. Extract mature human growth hormone (hGH) mRNA from pituitary gland;
2. Mature mRNA is reverse transcribed to form a single strand of hGH cDNA;
3. using reverse transcriptase;
4. mRNA degraded by RNase and;
5. complementary cDNA strand synthesised by DNA polymerase, forming double stranded hGH cDNA;
6. *HaeIII* cuts hGH cDNA into two fragments;
7. The longer fragment is used for the construction of recombinant plasmid;
8. The shorter fragment is replaced by artificial DNA molecule;
9. that reproduced the start of hGH gene;
10. and provide correct signals for translation in *E.coli*;
11. Modified hGH gene ligated into expression vector;
12. carrying lac promoter;
13. using DNA ligase;

Introduction of recombinant plasmid into *E.coli* host cell

14. Recombinant plasmid introduced into *E.coli* host cell;
15. Using calcium chloride and heat shock method;
16. *E. coli* colonies with recombinant plasmids are selected for;
17. Using selectable markers, Ref to antibiotics/synthesis of β -galactosidase;
18. Colony hybridization carried out to identify *E.coli* colonies with recombinant plasmid carrying hGH gene;

hGH protein production and isolation

19. Selected *E. coli* colonies are isolated and grown in suitable medium;
20. With lactose added to induce production of hGH;
21. hGH in growth medium is isolated for purification;

- (b) Explain how RFLP analysis facilitates the process of detecting sickle cell anaemia. [7]

Sickle cell anaemia

1. Sickle cell anaemia is an autosomal recessive disorder;
2. which is caused by base pair substitution mutation;
3. within a DdeI restriction site;
4. in the human β -globin gene;

Principle for analysis

5. DdeI recognises and cuts the specific DNA sequence 5' CTNAG 3';
6. in the mutant allele, there is a loss of DdeI restriction site;
7. prevents DdeI from recognising and cutting DNA at that particular restriction site;
8. DdeI restriction digest of normal and mutant allele will produce a restriction fragments of different number and length;

Techniques used and rationale

9. when gel electrophoresis is performed;
10. each allele will have its unique band pattern;
11. that help to distinguish the normal and sickle-cell alleles of the β -globin gene
12. Nucleic acid hybridisation;
13. reference to use of a radioactively-labelled, single-stranded DNA probe;
14. complementary to β -globin gene;

Analysis of observation

15. restriction fragment arising from the normal β -globin allele is shorter than that arising from the mutant β -globin allele;
16. reference to normal homozygotes having three band arising from normal allele only;;
17. reference to heterozygotes having more than 4 band, i.e. arising from both normal and mutant alleles;;
18. Individuals with sickle cell anemia have 2 bands;;

(c) Discuss the ethical concerns that arose from the human genome project. [5]

1. Fairness in/rights to the use/access of genetic information;
2. Legislation needed to ensure no discrimination on basis of genetic information;
3. In insurance/school/job entitlement/court (*any 1*):
 - a. Insurance - raising premiums based on genetic predisposition to disease;
 - b. Job entitlement - hiring and promoting based on genetic superiority rather than on merit;
 - c. Courts - judge a defendant based on a genetic predisposition to violence rather than the severity of the crime itself;
4. Using genetic information in reproductive decision making/ to decide the rights to reproduce;
5. Prenatal genetic testing could lead to genetic manipulation; Or decision to abort based on undesirable traits disclosed by test;
6. Lack of clear distinction between medical treatment and enhancement of humans;
7. Fairness in access to advanced genome technologies;
8. May widen worldwide inequities/unequal distribution of benefits;
9. E.g. designer drug;
10. Controversies related to commercialisation of products;
11. Due to award of patent for DNA sequences;
12. Which will limit accessibility and development of useful products using the information;
13. AVP;